

南京信息工程大学

Self-Supervised Representation Learning for Image Enhancement



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ABSTRACT

Image enhancement is a fundamental task in digital image processing, as noise and visual degradation can reduce clarity and negatively affect subsequent computer vision applications. While conventional filtering methods can suppress noise, they often blur edges and remove important image details. This paper therefore presents a self-supervised representation learning approach for image enhancement, based on a Masked Autoencoder-inspired U-Net model. This method learns without class labels by generating corrupted image views through the addition of Gaussian noise and random patch masking. Within this framework, the encoder learns useful visual representations from noisy and partially masked images, while the U-Net decoder reconstructs enhanced outputs by combining high-level semantic features with low-level spatial details. Experiments were conducted on the STL-10 dataset using 96×96 images. The model was trained for 20 epochs with a Gaussian noise standard deviation of 0.08, a masking ratio of 30%, and 8×8 image patches. Peak signal-to-noise ratio and structural similarity index were used to evaluate enhancement performance. The proposed model improved the average PSNR from 22.27 dB for noisy inputs to 28.89 dB for enhanced outputs, achieving a gain of 6.62 dB. The average SSIM increased from 0.6031 to 0.8875. Compared with a Gaussian blur baseline, the proposed method produced clearer images and preserved structural details more effectively. These results demonstrate the effectiveness of self-supervised representation learning as a deep learning-based solution for image enhancement, independent of class labels.

Keywords: Self-supervised learning; Image enhancement; Masked autoencoder; U-Net; Structural similarity index



摘要

图像增强是数字图像处理中的基础任务，因为噪声和图像质量下降会降低图像清晰度，并对后续的计算机视觉应用产生负面影响。虽然传统的滤波方法可以抑制噪声，但它们往往会模糊边缘并去除重要的图像细节。因此，本文提出了一种基于受掩码自编码器启发的 U-Net 模型的自监督表征学习方法，用于图像增强。该方法通过添加高斯噪声和随机补丁掩码来生成受损的图像视图，从而在没有类别标签的情况下进行学习。在此框架下，编码器从噪声和部分掩码的图像中学习有用的视觉表征，而 U-Net 解码器则通过结合高级语义特征与低级空间细节来重建增强后的输出。实验在 STL-10 数据集上使用 96×96 尺寸的图像进行。模型在高斯噪声标准差为 0.08、遮罩率为 30% 且采用 8×8 图像补丁的条件下，训练了 20 个 epoch。采用峰值信噪比 (PSNR) 和结构相似性指数 (SSIM) 来评估增强性能。所提出的模型将平均 PSNR 从噪声输入的 22.27 dB 提升至增强输出时的 28.89 dB，增益达 6.62 dB。平均 SSIM 从 0.6031 提升至 0.8875。与高斯模糊基线方法相比，所提方法生成的图像更清晰，且能更有效地保留结构细节。这些结果证明了自监督表征学习作为基于深度学习的图像增强解决方案的有效性，且该方法不依赖于类别标签。

关键词： 自监督学习；图像增强；掩码自编码器；U-Net；结构相似性指数



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Introduction

1.1 Background

Digital image processing is a major area of computer science and engineering focusing on the acquisition, enhancement, analysis, transformation and interpretation of digital images. Digital images are now widely used in modern applications such as medical diagnosis, remote sensing, autonomous driving, surveillance systems, industrial inspection, multimedia communication, intelligent transportation and general computer vision tasks. Image quality directly affects both human visual perception and the performance of automatic analysis systems. Therefore, improving image quality is an important research problem in digital image processing.

In real-world environments, images are often degraded during acquisition, transmission or storage. Common types of degradation include Gaussian noise, sensor noise, low illumination, motion blur, compression artefacts, low contrast and missing visual information. These degradations reduce image clarity and can hide important visual details, such as edges, textures, object boundaries and structural patterns. For instance, noise in an image can make identifying object shapes difficult, while excessive blur can obscure fine details. In computer vision systems, such degradation can reduce the accuracy of subsequent tasks such as classification, segmentation, detection, and recognition. Consequently, image enhancement is considered a fundamental and practical task in digital image processing.

The aim of image enhancement is to improve the visual appearance and structural quality of an image. An effective enhancement method should minimise unwanted noise while preserving meaningful details. Traditional methods include spatial filtering, histogram equalisation, sharpening, smoothing and frequency-domain processing. While these methods are computationally simple and easy to implement, they usually depend on manually designed rules. For instance, Gaussian blur can reduce random noise but may also blur object boundaries and remove texture details. Similarly, contrast enhancement methods may improve brightness, but they can also amplify noise. Therefore, classical enhancement techniques often struggle to balance noise suppression and detail preservation.

In recent years, however, artificial intelligence has significantly changed the field of digital image processing. Deep learning models can learn complex non-linear mappings from degraded images to enhanced images. Convolutional neural networks, autoencoders, U-Net architectures, generative adversarial networks and vision

transformers are widely used for tasks such as image restoration, denoising, super-resolution, segmentation, synthesis and enhancement. Unlike traditional filters, deep learning models can automatically learn useful image features from data. They can capture both low-level features, such as edges and textures, and high-level visual structures, such as object shapes and semantic regions. This makes deep learning models particularly well-suited to image enhancement tasks where both local details and global image structure are important.

However, many image enhancement methods based on deep learning use supervised learning. Supervised models typically require paired training data, consisting of a degraded image and its corresponding clean ground-truth image. In many real-world scenarios, collecting such paired data can be difficult or expensive. For instance, if an image is captured in noisy conditions, an exact clean version of the same image may not be available. This limitation has led to the development of self-supervised representation learning, whereby the model learns from the data itself without depending on human-provided labels.

Self-supervised representation learning is a strategy whereby a model is trained using automatically generated learning signals. Rather than using class labels, the model solves a pretext task created from the input data. In image processing, a common self-supervised strategy is to corrupt an image and train the model to reconstruct the original. Through this process, the model learns meaningful visual representations, such as edges, shapes, textures and object structures. Masked image modelling and masked autoencoders are important examples of this approach. In these methods, some parts of an image are obscured and the model learns to reconstruct the hidden areas using the remaining visible information. This forces the encoder to understand image structure rather than simply memorising pixel values.

This report focuses on the topic of self-supervised representation learning for image enhancement. Its main purpose is to investigate how a deep learning model can learn useful visual representations from corrupted images and then use these representations to produce enhanced outputs. The proposed method uses Gaussian noise and random patch masking to generate corrupted image views. A Masked Autoencoder-inspired U-Net model is then trained to reconstruct clean images from these corrupted inputs. The encoder acts as the representation of learning component, while the decoder performs image reconstruction and enhancement. The U-Net structure is employed due to the

fact that skip connections help to preserve spatial details, which is important for maintaining image clarity.

1.2 Problem Statement

Despite many years of research into image enhancement, achieving effective noise reduction while preserving important visual details remains challenging. Traditional filtering methods often have limitations because they apply fixed mathematical operations to all images. While these methods may perform well under simple degradation conditions, they are less effective when the image contains complex structures or different levels of noise. Strong smoothing can remove noise, but it may also blur the image and reduce structural information. Conversely, weak smoothing may preserve edges, but visible noise may remain. Therefore, striking the right balance between noise reduction and detail preservation is a central problem in image enhancement.

Deep learning methods are more effective because they can learn image restoration patterns directly from data. However, supervised deep learning models typically require paired clean and degraded images. Such data may not always be available for practical applications. This creates a need for a model that can learn useful image features independently, without relying on manual labels or large paired datasets.

This report addresses the problem of applying self-supervised representation learning to image enhancement. The model should learn from corrupted image views and produce cleaner images while preserving structural details. The objective is to achieve denoising and demonstrate that the encoder learns useful visual representations from the self-supervised pretext task. The proposed method therefore combines random patch masking, Gaussian noise corruption and a U-Net-based reconstruction model to support image enhancement.

1.3 Research Objectives

The main objective of this report is to design, implement and evaluate a self-supervised deep learning framework for image enhancement. The proposed framework uses a Masked Autoencoder-inspired U-Net architecture to learn visual representations from corrupted images and produce improved results.

The specific objectives of this work are as follows:

- 1) To study how self-supervised representation learning can be applied to a digital image enhancement task.
- 2) To implement a U-Net model inspired by Masked Autoencoders for image reconstruction and enhancement.
- 3) To use Gaussian noise and random patch masking as self-supervised corruption strategies.
- 4) The model will be trained without class labels to allow the encoder to learn useful image representations.
- 5) The proposed method will be evaluated using standard image quality metrics, including peak signal-to-noise ratio and structural similarity index.
- 6) Compare the proposed deep learning model with a traditional Gaussian blur baseline.
- 7) The model will be analyzed using qualitative and quantitative results, noise-level generalization and representation visualization.

1.4 Summary of the proposed study

The study uses the STL-10 dataset because it contains natural images with better visual resolution than very small datasets. All images are resized to 96×96 pixels. During training, Gaussian noise is added to clean images and random patch masking is applied to create corrupted image views. The model is then trained to reconstruct clean images from these corrupted inputs. During the evaluation stage, noisy images are fed directly into the trained model to generate enhanced outputs. The proposed model is then compared with both the noisy input and a Gaussian blur baseline.

Item	Description
Report title	Self-Supervised Representation Learning for Image Enhancement
Digital image processing task	Image enhancement and noise reduction
Artificial intelligence method	Deep learning-based self-supervised representation learning
Proposed model	Masked Autoencoder-inspired U-Net
Learning strategy	Self-supervised learning without class labels
Corruption methods	Gaussian noise and random patch masking
Dataset	STL-10 natural image dataset
Image resolution	96×96 pixels
Noise setting	Gaussian noise with $\sigma = 0.08$
Masking setting	30% random patch masking

Patch size	8×8 pixels
Training epochs	20 epochs
Baseline method	Gaussian blur filtering
Evaluation metrics	PSNR and SSIM
Main quantitative result	PSNR improved from 22.27 dB to 28.89 dB; SSIM improved from 0.6031 to 0.8875

Table 1 Summary of the Proposed Study

1.5 Main contributions

This report makes several significant contributions to the field of AI-based digital image processing. Firstly, it presents a self-supervised representation learning framework for image enhancement. Unlike traditional models, this one does not use class labels during training. Instead, it learns by reconstructing clean images from noisy and masked views of images. This makes the method suitable for situations where labelled data are limited or unavailable.

Secondly, the report combines masked image reconstruction with a U-Net-style encoder-decoder structure. The masked reconstruction task encourages the encoder to learn meaningful image representations, and the U-Net skip connections help to preserve spatial details during reconstruction. This combination is useful for image enhancement, as the model must reduce noise while maintaining important structures, such as edges, textures and object shapes.

Thirdly, the proposed method is evaluated using both qualitative and quantitative analyses. The experimental section includes visual enhancement results, a comparison with a Gaussian blur baseline, training loss curves, PSNR and SSIM evaluations, noise-level generalization and PCA visualization of encoder representations. Together, these provide a comprehensive evaluation of both enhancement performance and representation learning behavior.

Fourthly, the experimental results demonstrate that the proposed method significantly improves image quality. The average PSNR rises from 22.27 dB for noisy inputs to 28.89 dB for enhanced outputs. The average SSIM improves from 0.6031 to 0.8875. These results suggest that the self-supervised MAE-U-Net model is more effective than the Gaussian blur baseline at reducing noise and preserving structural details.

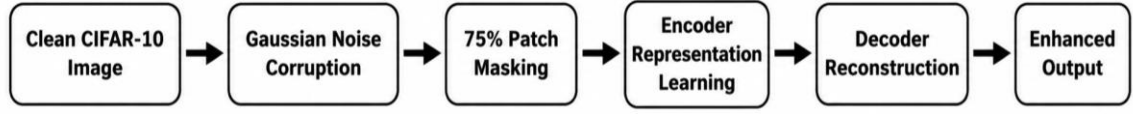


Figure 1 Self-Supervised Representation Learning Framework for Image Enhancement

Figure 1 illustrates the overall workflow of the proposed self-supervised image enhancement framework. First, the original image is degraded using Gaussian noise and random patch masking to generate a corrupted image. This corrupted image is then used as a self-supervised learning signal, meaning that the model does not require class labels during training. While the encoder learns visual representations from the corrupted input, the U-Net decoder reconstructs an enhanced image. This design links self-supervised representation learning with the practical task of image enhancement.

1.6 Importance of this study

This study is important because it connects self-supervised learning with the practical task of image enhancement. In many digital images processing applications, obtaining clean labels or paired clean images can be challenging. A self-supervised approach can reduce dependence on manually labelled data by creating training signals from the image itself. This makes the approach useful in real-world situations where data may be noisy, incomplete or poorly labelled.

The study also demonstrates the practical value of combining representation learning with image restoration. The encoder learns visual features from corrupted images, which are then used by the decoder to produce clearer outputs. This lends weight to the notion that self-supervised learning is not only useful for classification or pre-training but can also be applied directly to image enhancement. Consequently, the proposed approach is relevant to artificial intelligence and digital image processing alike.

1.7 Paper organisation

The remainder of this report is organised as follows. Section 2 presents related work on AI-based image enhancement, convolutional neural networks, U-Net architectures, masked autoencoders and self-supervised representation learning. Section 3 describes

the proposed methodology, including dataset preparation, preprocessing, Gaussian noise corruption, random patch masking, the MAE-U-Net architecture, loss functions and training configuration. Section 4 presents the experimental results and discussion, including qualitative enhancement results, a quantitative evaluation using PSNR and SSIM, a comparison with the Gaussian blur baseline, noise-level generalisation and a PCA visualisation of the learned representations. Section 5 concludes the report and discusses possible future improvements, including transformer-based masked autoencoders, larger datasets, modelling real-world noise, and perceptual loss functions.

2. Related work

Image enhancement has evolved from traditional filtering-based methods to modern, artificial intelligence-based restoration models. Earlier approaches relied mainly on manually designed mathematical operations, whereas recent methods use deep neural networks to learn image representations from data. As this report focuses on self-supervised representation learning for image enhancement, this section reviews the main research directions related to the proposed work. These include traditional image enhancement, convolutional neural network (CNN)-based restoration, U-Net architectures, self-supervised representation learning, masked autoencoders and evaluation methods for image enhancement.

2.1 Traditional Image Enhancement and Denoising Methods

Traditional image enhancement methods are based on mathematical transformations applied directly to pixel intensities or frequency components. Common methods include mean filtering, median filtering, Gaussian filtering, bilateral filtering, histogram equalisation, contrast stretching, sharpening filters and frequency-domain enhancement. These methods are widely used because they are simple, computationally efficient and easy to implement. Gaussian blur, for example, is a common smoothing method that reduces random noise by averaging neighbouring pixels with weights defined by a Gaussian kernel. Median filtering is often used for impulse noise, while histogram equalisation is used to improve global contrast.

Although classical methods are useful for many basic image processing tasks, they have significant limitations. For example, most traditional filters use fixed mathematical assumptions and are unable to adapt to complex image content. For example, a Gaussian filter may reduce visible noise, but it also smooths edges and removes fine details. Similarly, strong smoothing can improve uniform regions, but may blur object boundaries; conversely, weak smoothing may preserve structure, but leave noise in the output image. The trade-off between noise suppression and detail preservation is one of the main challenges in image enhancement.

Traditional methods also have a limited ability to learn from data. They do not automatically adapt to image category, texture, object structure or degradation type. In real-world images, noise is often spatially non-uniform and may occur alongside blur, low

contrast or compression artefacts. Therefore, a fixed filter may not be sufficient for high-quality enhancement. These limitations have encouraged the development of learning-based image enhancement methods, particularly deep learning models, which can learn complex restoration patterns from large datasets.

In this report, Gaussian blur is used as the baseline method. This is useful because it represents a classical, non-learning approach. By comparing the proposed model with Gaussian blur, it can be determined whether the self-supervised deep learning model outperforms traditional image enhancement methods.

2.2 Deep learning-based image enhancement

Deep learning has become the dominant approach in digital image processing, as neural networks can learn to recognise features directly from images. In image enhancement, deep learning models learn the mapping between degraded inputs and improved outputs. Convolutional neural networks are particularly effective in this context because convolutional layers can identify local spatial patterns, such as edges, textures, corners and object boundaries. These local features are important for image restoration and enhancement.

CNN-based enhancement methods typically comprise three main components: feature extraction, non-linear transformation and image reconstruction. The feature extraction component learns low- and mid-level features from the degraded image. The nonlinear transformation component processes these features through multiple convolutional layers. The reconstruction stage generates enhanced output image. Unlike traditional filtering methods, CNNs learn their own filters during training, enabling them to adapt to different image structures and degradation patterns.

Deep learning-based enhancement has been applied to many related tasks, including image denoising, super-resolution, deblurring, low-light enhancement and image inpainting. These tasks are closely related because they all require the model to produce a visually improved image from degraded input. In many cases, deep learning models outperform traditional filters because they can preserve semantic structure while reducing noise.

However, many deep learning enhancement models are trained using a supervised approach. Supervised learning requires paired degraded and clean images. This means that each noisy image must have a corresponding clean 'ground truth' image. In controlled experiments, such pairs can be generated synthetically by adding noise to clean images. In real-world

applications, however, collecting exact clean-degraded image pairs is difficult. This creates a limitation for supervised enhancement methods.

The proposed work addresses this issue by using a self-supervised learning strategy. Rather than relying on class labels, the model learns from corrupted image views generated from the image itself. This makes the method more aligned with modern representation learning, in which models learn useful visual features through reconstruction-based tasks.

2.3 Autoencoders and reconstruction-based learning

Autoencoders are neural network models designed to learn compressed representations of input data. A basic autoencoder consists of an encoder and a decoder. The encoder maps the input image into a latent representation, and the decoder reconstructs the image from that representation. Autoencoders are useful in image enhancement because they can learn to reconstruct clean images from degraded inputs.

Denoising autoencoders build on this concept by training the model to reconstruct a clean image from a distorted version. In this setup, the input image is deliberately distorted by noise or masking, and the model learns to recreate the original image. This process encourages the encoder to identify robust features rather than simply copying the input. Denoising autoencoders are closely related to self-supervised learning, since the training signal can be derived from the image itself.

However, simple autoencoders may produce blurry outputs because the bottleneck representation can remove fine spatial information. If the encoder compresses the image too strongly, details such as edges and textures may be lost. This issue is particularly significant in image enhancement, where visual quality hinges on preserving both global structure and local detail.

To mitigate this issue, many image restoration models employ encoder-decoder networks with skip connections. These connections allow the decoder to access feature maps from earlier encoder layers. This enables the model to recover fine details and improves reconstruction quality. The proposed method follows this approach, using a U-Net-style architecture.

2.4 U-Net architecture for image restoration

Originally developed for biomedical image segmentation, U-Net is a widely used encoder-decoder architecture that has also become popular for image restoration and enhancement tasks. The main strength of the U-Net architecture is its skip connection design. In a U-Net,



the encoder gradually reduces the spatial resolution while increasing the feature depth. This enables the model to learn high-level features. The decoder then reconstructs the image by increasing the spatial resolution. Skip connections transfer feature maps from the encoder layers to the corresponding decoder layers.

U-Net is useful for image enhancement because it can combine two types of information. The deeper layers capture global image structure and semantic context, while the skip connections preserve low-level details such as edges, contours and textures. This is important because image enhancement should remove noise while maintaining the natural appearance of objects.

Compared with a plain autoencoder, U-Net usually produces clearer outputs because it does not force all image information to pass through a single bottleneck. Instead, the decoder receives both high-level representations and low-level spatial details. This reduces excessive smoothing and improves visual reconstruction.

In this report, the proposed model uses a U-Net structure inspired by Masked Autoencoders. The encoder acts as the representation learning component and the decoder is responsible for enhancement. This design is relevant to the topic of the report because it combines self-supervised representation learning with practical image enhancement.

2.5 Self-supervised representation learning

Self-supervised representation learning is an important area of research in artificial intelligence. The basic concept involves training a model without manually assigned labels by designing a pretext task using the data itself. The model learns useful features by solving this task. Once trained, these representations can support downstream tasks such as classification, detection, segmentation or restoration.

In computer vision, self-supervised learning has been studied using various approaches. Some methods use contrastive learning, whereby the model learns to bring similar image views closer together in feature space while pushing different images apart. Other methods use reconstruction-based learning, where the model learns to restore missing or corrupted parts of an image. Examples of reconstruction-based tasks include image inpainting, colourisation, denoising and masked image modelling.

Reconstruction-based self-supervised learning is especially relevant for image enhancement. The model can be trained by corrupting an image and asking the model to recreate the original version. This process does not require class labels because the image itself provides the learning signal. During training, the model must learn about the structure,

texture and objects in the image in order to recover the original content. This makes self-supervised learning suitable for enhancement tasks where labelled data may be limited.

The proposed work builds on this concept by generating corrupted image views via the addition of Gaussian noise and random patch masking. The model then learns representations from these corrupted views and uses them to reconstruct enhanced outputs. This approach is consistent with the concept of self-supervised representation learning, since the model learns from the image structure itself rather than from external labels.

2.6 Masked Autoencoders and Masked Image Modelling

Masked image modelling is a self-supervised learning method in which parts of an image are hidden and the model is trained to reconstruct the missing information. Masked autoencoders follow this principle by dividing an image into patches, masking some of them, and training an encoder-decoder model to reconstruct the full image. This forces the model to understand the image's overall structure because it must infer the missing regions from the visible context.

Masked autoencoders have become influential because they provide a simple yet effective means of learning visual representations. Rather than requiring class labels, the model learns by solving a reconstruction problem. The encoder learns to represent visible image information compactly, while the decoder reconstructs the missing regions. This learning strategy is useful for pre-training and can also be adapted for image restoration tasks.

In the proposed work, masked image modelling is adapted for image enhancement. First, the input image is corrupted using Gaussian noise, and then random patch masking is applied. This creates a more challenging self-supervised learning task, as the model must learn to handle both noise and missing image regions. The model is then trained to reconstruct clean images from these corrupted views. Consequently, the encoder learns representations that are useful for enhancement.

The proposed method is more than just a standard masked autoencoder. Instead, it combines MAE-style masking with a U-Net reconstruction architecture. This combination is important because standard autoencoder reconstruction can produce blurry results, particularly when the masking ratio is high. The U-Net decoder improves reconstruction by preserving spatial information via skip connections.

2.7 Loss functions for image enhancement

Loss functions are important in image enhancement as they determine how the model learns to reconstruct the output image. A simple pixel-level loss function, such as mean squared error (MSE), encourages the model to minimise the average difference between the predicted and target pixels. MSE is useful for improving PSNR, but it can produce overly smooth images as it strongly penalises large pixel errors. Mean absolute error and Charbonnier loss are often used as alternatives because they are less sensitive to outliers and can preserve sharper structures.

Structural similarity loss is also important because image quality is not solely determined by pixel-wise error. Human perception is sensitive to structural information, such as edges, contrast and texture. The structural similarity index measures image similarity based on luminance, contrast, and structure. Therefore, incorporating SSIM-related loss functions can encourage the model to preserve visual structure more effectively.

The proposed method uses a combined loss strategy that incorporates both reconstruction-based loss and structural similarity information. This enables the model to enhance both pixel-level accuracy and perceptual structure. The final evaluation uses PSNR and SSIM, which are standard metrics for assessing the quality of images in restoration and enhancement tasks.

2.8 Evaluation metrics for image enhancement

Image enhancement methods are typically evaluated using qualitative and quantitative analyses. Qualitative analysis involves visually comparing enhanced outputs with noisy inputs and reference images. This is important because visual quality is the primary objective of image enhancement. However, visual inspection alone is subjective, so quantitative metrics are also required.

Peak signal-to-noise ratio (PSNR) is a widely used metric for measuring image reconstruction quality. A higher PSNR value usually indicates that the enhanced image more closely resembles the clean reference image at a pixel level. Another important metric is the structural similarity index, which measures structural similarity between images. SSIM is useful because it considers luminance, contrast and structural information. A higher SSIM value indicates better preservation of image structure.

In this report, PSNR and SSIM are used to evaluate the proposed MAE-U-Net model. The model is compared with results from noisy input and a Gaussian blur baseline. This

comparison shows whether the proposed deep learning method improves image quality more effectively than traditional smoothing techniques.

2.9 Research Gap and Motivation for the Proposed Method

The literature shows that, while traditional enhancement methods are simple, they are often limited in their ability to preserve fine details. Although deep learning methods improve the quality of enhancement, supervised models often require paired clean and degraded data. Self-supervised learning reduces the need for labels by creating learning signals directly from the input image. Masked autoencoders are an effective way of learning image representations through reconstruction; however, standard reconstruction models may still produce blurred outputs if spatial detail is not preserved.

This creates a research gap for a method that can combine self-supervised representation learning with suitable architecture for detail-preserving image enhancement. The proposed work addresses this issue by combining Gaussian noise corruption, random patch masking and a U-Net-based reconstruction model. The masking strategy supports self-supervised representation learning, while the U-Net structure improves spatial reconstruction. The result is a model that learns without class labels and produces enhanced images with improved PSNR and SSIM scores.

Table 2 summarises the reviewed approaches and presents a taxonomy of major image enhancement methods, including traditional filtering, convolutional neural network (CNN)-based restoration, autoencoder-based reconstruction, U-Net-based enhancement, self-supervised learning and masked autoencoder methods.

Category	Representative idea	Main strengths	Main limitations	Relevance to this report
Traditional filtering	Gaussian blur, median filtering, histogram equalization	Simple, fast, easy to implement	Limited adaptability; may blur edges and textures	Used as the classical baseline
CNN-based restoration	Learns mapping from degraded image to clean image	Learns adaptive filters from data	Often requires paired clean-degraded data	Provides foundation for deep image enhancement
Autoencoder-based reconstruction	Encodes degraded input and decodes	Learns compact image representation	May produce blurred outputs due to bottleneck compression	Related to reconstruction-based learning

	reconstructed output			
U-Net-based enhancement	Uses encoder-decoder with skip connections	Preserves spatial details and object boundaries	Can still require supervised training	Used as the main architecture style
Self-supervised learning	Learns from automatically generated pretext tasks	Reduces dependence on labels	Requires carefully designed corruption/reconstruction tasks	Core learning strategy of this report
Masked autoencoder	Reconstructs hidden image patches from visible context	Learning meaningful visual representations	High masking can reduce visual sharpness	Inspires the proposed masking strategy
Proposed MAE-U-Net	Combines noise corruption, patch masking, and U-Net reconstruction	Learning without labels and improves image quality	Evaluated mainly on synthetic Gaussian noise	Final implemented method in this report

Table 2 Taxonomy of Image Enhancement Approaches

Based on the reviewed literature, it is evident that several limitations remain in existing enhancement methods. Table 3 summarises these research gaps and explains how the proposed MAE-U-Net framework addresses them.

Existing issue	Effect on image enhancement	Proposed solution in this report
Traditional filters are fixed and non-adaptive	Noise may be reduced, but edges and textures may be blurred	Use a data-driven deep learning model
Supervised methods require paired clean-degraded images	Paired datasets may be difficult to obtain in practice	Use self-supervised reconstruction without class labels
Simple autoencoders may lose spatial details	Enhanced outputs may become overly smooth	Use U-Net skip connections for detail preservation
Masked reconstruction alone may be difficult	Heavy masking can reduce output clarity	Use moderate masking and direct noisy-image enhancement during evaluation
Pixel-level losses may not fully preserve structure	Image may have high pixel accuracy but weak perceptual quality	Evaluate and optimize using both PSNR and SSIM-based structural quality

Table 3 Research Gap Addressed by the Proposed Method

3. Methodology

3.1 Overview of the proposed method

This section describes the proposed self-supervised representation learning framework for image enhancement. It is designed to learn useful visual representations from corrupt image views, which are then used to reconstruct enhanced images. The model is based on a U-Net architecture inspired by Masked Autoencoders. It combines two key concepts: masked reconstruction for self-supervised learning and U-Net-based image reconstruction to preserve detail.

The main workflow of the proposed method consists of five stages. First, clean images are loaded from the STL-10 dataset and resized to 96×96 pixels. Second, Gaussian noise is added to the images to simulate degraded visual conditions. Third, random patch masking is applied to create a self-supervised reconstruction task. Fourthly, the corrupted image is passed through the MAE-U-Net model, where the encoder learns visual representations and the decoder reconstructs the enhanced image. Finally, the output is evaluated using PSNR and SSIM.

The proposed framework does not use class labels during training. Although the STL-10 dataset contains class labels, these are not used for optimisation. Instead, the learning signal is created from the image itself by corrupting the input and asking the model to reconstruct a clean version of the image. This is why the method is considered self-supervised. The encoder learns to recognise image structures from noisy and masked views, and the decoder uses these features to generate improved outputs.

3.2 Dataset and preprocessing

The STL-10 dataset was used in this experiment because its images have a higher visual resolution than those in smaller datasets, such as CIFAR-10. STL-10 images are ideal for image enhancement experiments as the objects and backgrounds are highly recognisable. For this study, the images were resized to 96×96 pixels to provide a better balance between image clarity and computational efficiency.

The dataset contains 5,000 training images and 8,000 test images. These images belong to categories of natural objects such as aeroplane, bird, car, cat, deer, dog, horse, monkey, ship and truck. However, the class labels were not used during training as the aim of this study

is self-supervised representation learning. The model only learns from image reconstruction signals.

During preprocessing, each image was converted into a tensor and normalised to the range $[0, 1]$. Random horizontal flipping was applied to the training set to improve data variation and reduce overfitting. The test images were resized and converted to tensors without random augmentation to ensure consistent evaluation.

Item	Description
Dataset	STL-10
Image type	Natural RGB images
Number of training images	5,000
Number of testing images	8,000
Image resolution used	96×96 pixels
Number of channels	3 RGB channels
Training augmentation	Random horizontal flipping
Tensor range	$[0, 1]$
Class labels used during training	No
Learning type	Self-supervised reconstruction

Table 4 Dataset and Preprocessing Configuration

Using STL-10 improves the visual quality of the experimental results, as the images are larger and clearer than those in the 32×32 dataset. This is important for the final report, as image enhancement should be evaluated through both numerical metrics and visible image quality.

3.3 Self-supervised corruption strategy

The proposed method uses a self-supervised corruption strategy to generate the training task. Rather than using manually assigned class labels, the model is given a corrupted image and learns to reconstruct the original, clean version. Two types of corruption are applied: Gaussian noise and random patch masking.

3.3.1 Gaussian noise corruption

Gaussian noise is added to the clean image to simulate noisy image acquisition. This type of noise is commonly used in image restoration research because it represents random disturbance at the pixel level. In this experiment, the standard deviation of the Gaussian noise was set to $\sigma = 0.08$. This value creates visible degradation while still making the object structure recognisable.

The noisy image is generated as:

```
[
I_{noisy} = clamp(I_{clean} + \epsilon, 0, 1)
]
```

where (I_{clean}) is the original image, (ϵ) is Gaussian noise, and the clamp operation ensures that pixel values remain between 0 and 1.

Gaussian Noise Corruption

```
def add_gaussian_noise(x, std=0.08):
    noisy = x + torch.randn_like(x) * std
    return torch.clamp(noisy, 0.0, 1.0)
```

This noise corruption step allows the model to learn denoising behavior. During training, the model receives noisy images and learns to reconstruct the clean version.

3.3.2 Random patch masking

In addition to Gaussian noise, a masked reconstruction task inspired by masked autoencoders is created using random patch masking. The image is divided into small 8×8 pixel patches. A fixed percentage of these patches is randomly hidden. In this study, the masking ratio was set to 30%. This means that 30% of the image patches are removed from the input view.

Patch masking encourages the model to learn contextual information from the visible parts of the image. The encoder cannot simply copy the input because some regions are missing. Instead, it must learn meaningful visual features to help it reconstruct the hidden regions. This makes the training process self-supervised and supports representation learning.

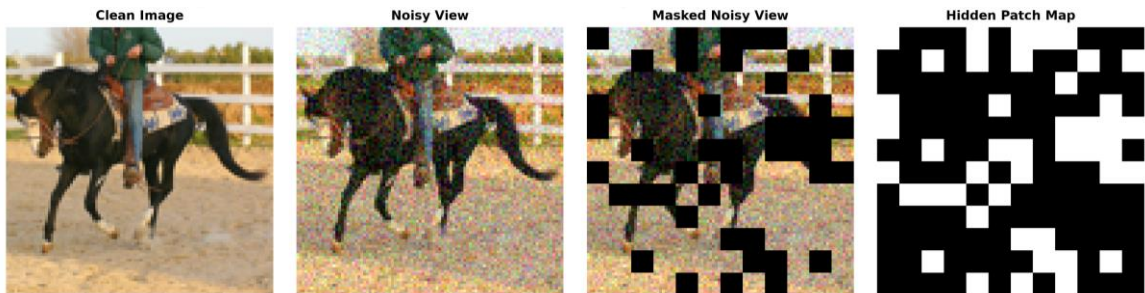


Figure 2 Noise Corruption and Patch Masking Used for Self-Supervised Learning

Figure 2 should be inserted here because it illustrates the two corruption operations used in the proposed method. It shows the clean image, the noisy view, the masked noisy view and the hidden patch map. This helps the reader to understand how the self-supervised reconstruction task is created.

Random Patch Masking

```
def mask_patches(x, patch_size = 8, mask_ratio = 0.30):
    B, C, H, W = x.shape
    ph = H // patch_size
    pw = W // patch_size
    total_patches = ph * pw
    num_visible = int(total_patches * (1.0 - mask_ratio))

    random_scores = torch.rand(B, total_patches,
                                device=x.device)
    visible_indices =
random_scores.argsort(dim=1)[: , :num_visible]

    patch_mask = torch.zeros(B, total_patches,
                                device=x.device)
    patch_mask.scatter_(1, visible_indices, 1.0)

    patch_mask = patch_mask.view(B, 1, ph, pw)
    visible_mask =
patch_mask.repeat_interleave(patch_size, dim =
2).repeat_interleave(patch_size, dim = 3)
    hidden_mask = 1.0 - visible_mask
```

Combining Gaussian noise and patch masking makes the task more challenging than simple denoising. The model must learn to suppress noise and infer missing image structure from surrounding visible regions.

3.4 Proposed MAE-U-Net architecture

The proposed model is a masked autoencoder-inspired U-Net. The architecture contains an encoder, a bottleneck and a decoder. The encoder learns visual representations from corrupted image inputs. The bottleneck stores compressed, high-level features. The decoder then uses these features and skip connections from the encoder to reconstruct the enhanced image.

The model adheres to the fundamental principle of masked autoencoders in that it learns from masked and corrupted image data. However, rather than using a simple autoencoder decoder, this work employs a U-Net-style decoder. This is important because image enhancement requires the preservation of spatial details. A standard autoencoder may produce overly smoothed results because the image must pass through a compressed bottleneck. U-Net skip connections mitigate this issue by transferring low-level spatial information directly from the encoder to the decoder.

3.4.1 Encoder:

The encoder comprises several convolutional blocks. Each block extracts visual features at a different scale. In the early layers, the model learns low-level features such as edges, textures and local contrast. In the deeper layers, the model learns higher-level features, such as object shapes and broader structural patterns. Max-pooling is used to reduce spatial resolution and increase the receptive field.

The encoder is the main component responsible for representation learning in the model. During training, it receives noisy and masked image views. In order to reconstruct the clean image, the encoder must learn meaningful information about image structure. This learned representation is then used by the decoder to enhance the image.

3.4.2 Bottleneck

The bottleneck is the deepest part of the network. It contains the most compressed representation of the input image. This layer captures high-level visual information following several stages of feature extraction. In self-supervised representation learning, the bottleneck is important because it stores the compact features that have been learned from the reconstruction task.

An effective bottleneck representation should contain sufficient information to reconstruct the primary content of an image while disregarding irrelevant noise. This is one reason why denoising and masked reconstruction are useful self-supervised tasks. They encourage the model to learn stable features rather than memorising random noise.

3.4.3 Decoder and skip connections

The decoder uses the learned representation to reconstruct the enhanced image. It uses transposed convolution layers to gradually increase the spatial resolution. At each reconstruction stage, the decoder receives feature maps from the corresponding encoder stage via skip connections. These skip connections help to recover spatial details that may be lost during downsampling.

The final output layer uses a sigmoid activation function to ensure that the predicted image values remain within the range $[0, 1]$. The output image is the same size as the input image: $96 \times 96 \times 3$.

Component	Operation	Output role
Input layer	96×96 RGB image	Receives noisy or masked image

Encoder block 1	Convolution + BatchNorm + ReLU	Extracts low-level edges and textures
Encoder block 2	Downsampling + convolution	Learns mid-level spatial features
Encoder block 3	Downsampling + convolution	Learns object-level structures
Encoder block 4	Downsampling + convolution	Learns high-level representations
Bottleneck	Deep convolutional representation	Stores compact visual features
Decoder block 4	Upsampling + skip connection	Reconstructs high-level spatial structure
Decoder block 3	Upsampling + skip connection	Restores mid-level image details
Decoder block 2	Upsampling + skip connection	Recovers local visual patterns
Decoder block 1	Upsampling + skip connection	Refines final image details
Output layer	Convolution + sigmoid	Produces enhanced RGB image

Table 5 Proposed MAE-U-Net Architecture Configuration

This architecture is well-suited to the selected topic as it supports both representation learning and image enhancement. The encoder learns useful visual representations and the decoder reconstructs the enhanced output image.

3.5 Loss Function Design

The loss function plays a key role in the proposed method by guiding the model to produce high-quality enhanced images. However, a single pixel-level loss function may not be sufficient, as image enhancement requires both numerical accuracy and structural preservation. Therefore, the proposed method uses a combined loss function.

The training objective comprises three main components: denoising reconstruction loss, masked reconstruction loss and hidden-region reconstruction loss.

The overall loss can be written as follows:

$$[\begin{aligned} L_{\text{total}} &= 0.60L_{\text{denoise}} + 0.30L_{\text{masked}} + \\ &0.10L_{\text{hidden}} \end{aligned}]$$

where (L_{denoise}) measures reconstruction from noisy input, (L_{masked}) measures reconstruction from masked noisy input, and (L_{hidden}) focuses on masked regions.

The image reconstruction loss combines Charbonnier loss, mean squared error, and SSIM-based structural loss:

```
[
L_{image} = 0.55L_{charbonnier} + 0.25L_{mse} +
0.20L_{ssim}
]
```

Charbonnier loss is used because it is more stable than mean squared error and less sensitive to differences between pixels. MSE supports pixel-level reconstruction and improves PSNR. SSIM-based loss encourages the model to preserve structural information, such as edges, contrast and object boundaries.

This combined loss design enables the model to produce outputs that are numerically close to the clean image and visually meaningful.

Loss component	Purpose	Contribution
Charbonnier loss	Stable pixel-level reconstruction	Reduces sensitivity to outliers
MSE loss	Pixel-wise error minimization	Supports higher PSNR
SSIM loss	Structural preservation	Supports higher visual similarity
Masked reconstruction loss	Learns from masked corrupted views	Supports self-supervised representation learning
Hidden-region loss	Focuses on missing patches	Encourages contextual reconstruction

Table 6 Loss Function Components

3.6 Training configuration:

The model was trained for 20 epochs using the AdamW optimiser. A cosine annealing learning rate scheduler was employed to adjust the learning rate gradually during training. The batch size was set to 16 to strike a balance between memory usage and training stability. When available, the model was trained using a GPU.

During each training step, the model receives both noisy and masked noisy images. The former is used for direct enhancement learning and the latter for self-supervised masked reconstruction. Both outputs were then compared with the clean image. This dual-branch training strategy enables the model to learn denoising and representation learning simultaneously.

Parameter	Value
Dataset	STL-10
Image size	96 × 96
Batch size	16
Number of epochs	20
Optimizer	AdamW
Learning rate	0.001
Scheduler	Cosine annealing

Gaussian noise standard deviation	0.08
Patch size	8×8
Masking ratio	30%
Training labels used	No
Evaluation metrics	PSNR and SSIM
Baseline	Gaussian blur

Table 7 Training Hyperparameter Configuration

Training Step Logic

```

clean = imgs.to(device)

noisy = add_gaussian_noise(clean, std=0.08)
masked, visible_mask, hidden_mask = mask_patches(noisy)

pred_noisy = model(noisy)
pred_masked = model(masked)

loss, denoise_loss, masked_loss, hidden_loss = criterion(
    pred_noisy,
    pred_masked,
    clean,
    hidden_mask
)
optimizer.zero_grad()
loss.backward()
optimizer.step()

```

This training process is self-supervised, as the model does not use object class labels. After the input is corrupted by noise and masking, the clean image acts as the reconstruction target.

3.7 Evaluation Strategy

The model was evaluated using qualitative and quantitative methods. Qualitative evaluation involved visually comparing clean images, noisy images, masked views, Gaussian blur outputs and enhanced outputs. This visual comparison shows whether the proposed model improves image clarity and preserves object structure.

Quantitative evaluation was performed using PSNR and SSIM. PSNR measures pixel-level reconstruction quality. A higher PSNR indicates that the enhanced image is closer to the clean reference image. SSIM measures the structural similarity between the enhanced and clean images. A higher SSIM value indicates better preservation of luminance, contrast and structural information.

The proposed model was also compared with a Gaussian blur baseline. This comparison is important because Gaussian blur is a traditional noise reduction method. If the proposed

model achieves a higher PSNR and SSIM than Gaussian blur, this demonstrates that the deep learning-based approach provides stronger enhancement than classical filtering methods.

During evaluation, the model receives noisy images directly. Patch masking is mainly used during training to support self-supervised representation learning. This design is important because the final practical task is to enhance noisy images, not reconstructing black-masked images.

Final Enhancement Evaluation

```
with torch.no_grad():  
    noisy = add_gaussian_noise(clean, std=0.08)  
    enhanced = model(noisy)
```

4. Experiments and discussion

4.1 Experimental setup

This section presents an evaluation of the proposed self-supervised MAE-U-Net model for image enhancement. The experiments were designed to verify whether the model can learn useful visual representations from corrupted image views and use these to reconstruct enhanced images. The evaluation includes a qualitative visual comparison, a quantitative measurement using PSNR and SSIM, a comparison with a traditional Gaussian blur baseline, noise-level generalisation and a visualisation of the learned encoder representations.

The proposed method was implemented using Python and PyTorch. The model was trained on the STL-10 dataset using 96×96 pixel images. During training, Gaussian noise and random patch masking were applied to generate corrupted image views. The model learnt to reconstruct clean images from these corrupted inputs without using class labels. During evaluation, noisy images were passed directly into the trained model to generate enhanced outputs.

The experiment aimed to determine whether the proposed self-supervised learning strategy could improve image quality compared to noisy inputs and a classical non-learning baseline. Gaussian blur was selected as the baseline because it is a common, traditional method of noise reduction. However, Gaussian blur often blurs edges and textures. Therefore, comparing the proposed model with Gaussian blur is a useful way to evaluate whether the deep learning model can reduce noise while preserving structure more effectively.

Item	Experimental setting
Dataset	STL-10
Image resolution	96×96 pixels
Image channels	3 RGB channels
Training images	5,000
Testing images	8,000
Main evaluation images	500
Model	Self-Supervised MAE-U-Net
Training strategy	Self-supervised reconstruction
Class labels used	No
Noise type	Gaussian noise
Noise standard deviation	$\sigma = 0.08$
Patch size	8×8 pixels
Masking ratio	30%
Training epochs	20

Batch size	16
Optimizer	AdamW
Learning rate	0.001
Baseline method	Gaussian blur
Evaluation metrics	PSNR and SSIM

Table 8 Experimental Configuration

The experimental setup was chosen to strike a balance between image clarity, computational cost and model performance. STL-10 was chosen over smaller image datasets because its 96×96 images provide clearer visual details for enhancement evaluation. A moderate level of Gaussian noise ($\sigma = 0.08$) was applied to the input images to degrade them visibly yet still make them recognisable. The 30% patch masking ratio was chosen to create a meaningful self-supervised reconstruction task that was still possible to complete.

4.2 Evaluation metrics

The performance of the proposed model was evaluated using two standard image quality metrics: peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM).

Peak signal-to-noise ratio measures the similarity of pixels between the enhanced image and the clean reference image. It is based on the mean squared error between the two images. A higher PSNR value indicates that the reconstructed image is closer to the clean image at the pixel level. PSNR is commonly used to measure reconstruction accuracy in image restoration and enhancement tasks.

PSNR is calculated as follows:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

Where MAX_I is the maximum possible pixel value and MSE is the mean squared error between the clean and enhanced images. As the images in this experiment were normalised to the range $[0, 1]$, MAX_I was set to 1.

The structural similarity index measures the structural similarity between two images. Unlike PSNR, SSIM considers luminance, contrast and structural information. This makes SSIM useful for evaluating perceptual image quality. A higher SSIM value indicates that the enhanced image retains more structural information from the clean reference image.

SSIM can be expressed as follows:

$$SSIM(x,y)=\frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$$

Where x and y represent two images, μ_x and μ_y are their mean intensities, σ_x^2 and σ_y^2 are their variances, and σ_{xy} is the covariance between the two images.

In this experiment, both PSNR and SSIM are important. PSNR evaluates the quality of numerical reconstruction, while SSIM evaluates structural preservation. An effective image enhancement model should improve both metrics.

Metric	Full name	Purpose	Better value
PSNR	Peak Signal-to-Noise Ratio	Measures pixel-level reconstruction accuracy	Higher
SSIM	Structural Similarity Index	Measures structural and perceptual similarity	Higher

Table 9 Evaluation Metrics Used in the Experiment

4.3 Training performance

The proposed MAE-U-Net model was trained for 20 epochs. During this process, the model was presented with two corrupted views of each image: a noisy image and a masked noisy image. The noisy image was used for direct denoising reconstruction and the masked noisy image for self-supervised masked reconstruction. The predicted outputs from both processes were then compared with the clean image. This training process enabled the model to learn image enhancement and representation learning simultaneously.

The training loss decreased consistently across epochs. This indicates that the model gradually learned to reconstruct clean images from corrupted inputs. Initially, the model had limited ability to remove noise and reconstruct image structure. As training continued, however, the model improved its ability to preserve object shapes, suppress noise and generate clearer outputs.

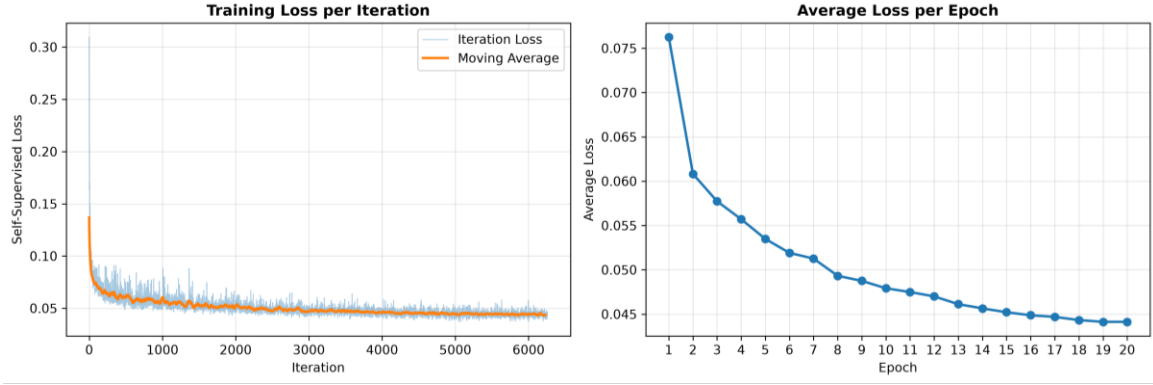


Figure 3 Training curves of the self-supervised MAE-U-Net model

Figure 3 shows the training loss curves of the proposed model. The left plot shows the loss at each iteration, while the right plot shows the average loss per epoch. The downward trend confirms that the self-supervised reconstruction task was learnt effectively. The average loss decreased from around 0.076 at the first epoch to approximately 0.044 at the twentieth epoch. This consistent reduction in loss suggests that the model converged properly without exhibiting unstable training behavior.

The training curve is important because it shows that the model is learning from the self-supervised task, rather than producing random results. Loss reduction also supports the subsequent qualitative and quantitative results. As the model was trained without class labels, the reduction in loss demonstrates that the reconstruction signal derived from corrupted image views was sufficient for learning useful visual features.

4.4 Qualitative enhancement results

A qualitative evaluation was conducted by comparing clean images, noisy inputs, masked views and enhanced outputs. This visual comparison helps to determine whether the proposed model improves the appearance of images in a way that is visible to human observers. Since image enhancement is partly a visual task, qualitative results are necessary in addition to numerical metrics.

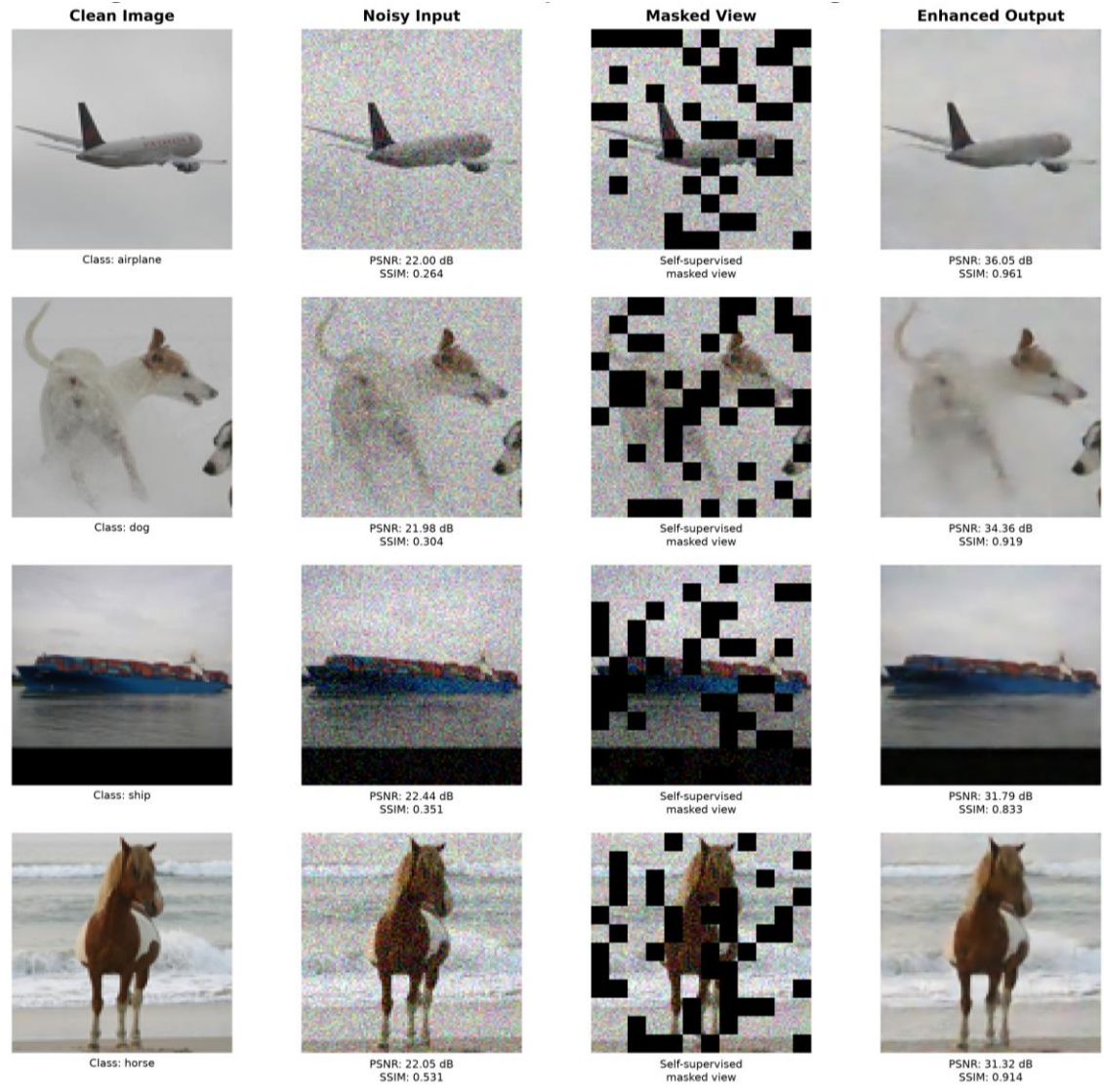


Figure 4 Qualitative Results of Self-Supervised MAE-U-Net Image Enhancement

Figure 4 shows representative enhancement results produced by the proposed model. Clean images provide the reference appearance, while noisy inputs show degraded images after Gaussian noise corruption. The masked views demonstrate the self-supervised reconstruction process employed during training. The final results generated by the trained MAE-U-Net model are shown in the enhanced outputs.

These visual results demonstrate that the proposed model effectively reduces noise while maintaining object structure. Compared with the noisy inputs, the enhanced outputs appear smoother and cleaner. Important visual structures, such as object boundaries and major shapes, are also preserved. This is important because an enhancement model should not simply blur the image. It should remove noise while retaining useful image details.

The results also demonstrate the advantages of using a U-Net-style architecture. The enhanced images are clearer than earlier simple autoencoder outputs because skip connections preserve spatial information. This reduces the excessive blurring that often occurs in basic reconstruction models. While the encoder learns high-level visual representations, the decoder uses these high-level features, as well as low-level spatial information, to reconstruct the image.

While the enhanced images are not identical to the clean references, they resemble the clean images much more closely than the noisy inputs. Some regions may still exhibit minor smoothing, which is common in image restoration models trained with reconstruction-based losses. Nevertheless, the overall visual quality is significantly improved.

4.5 Comparison with Gaussian Blur Baseline

To evaluate the effectiveness of the proposed model, we compared the results with those of a traditional Gaussian blur baseline. Gaussian blur is a common method for reducing noise. It smooths an image using a Gaussian kernel. While this suppresses random noise, it often removes edge details and weakens object boundaries. Therefore, it is useful for testing whether the proposed deep learning model can achieve better noise reduction and structure preservation.



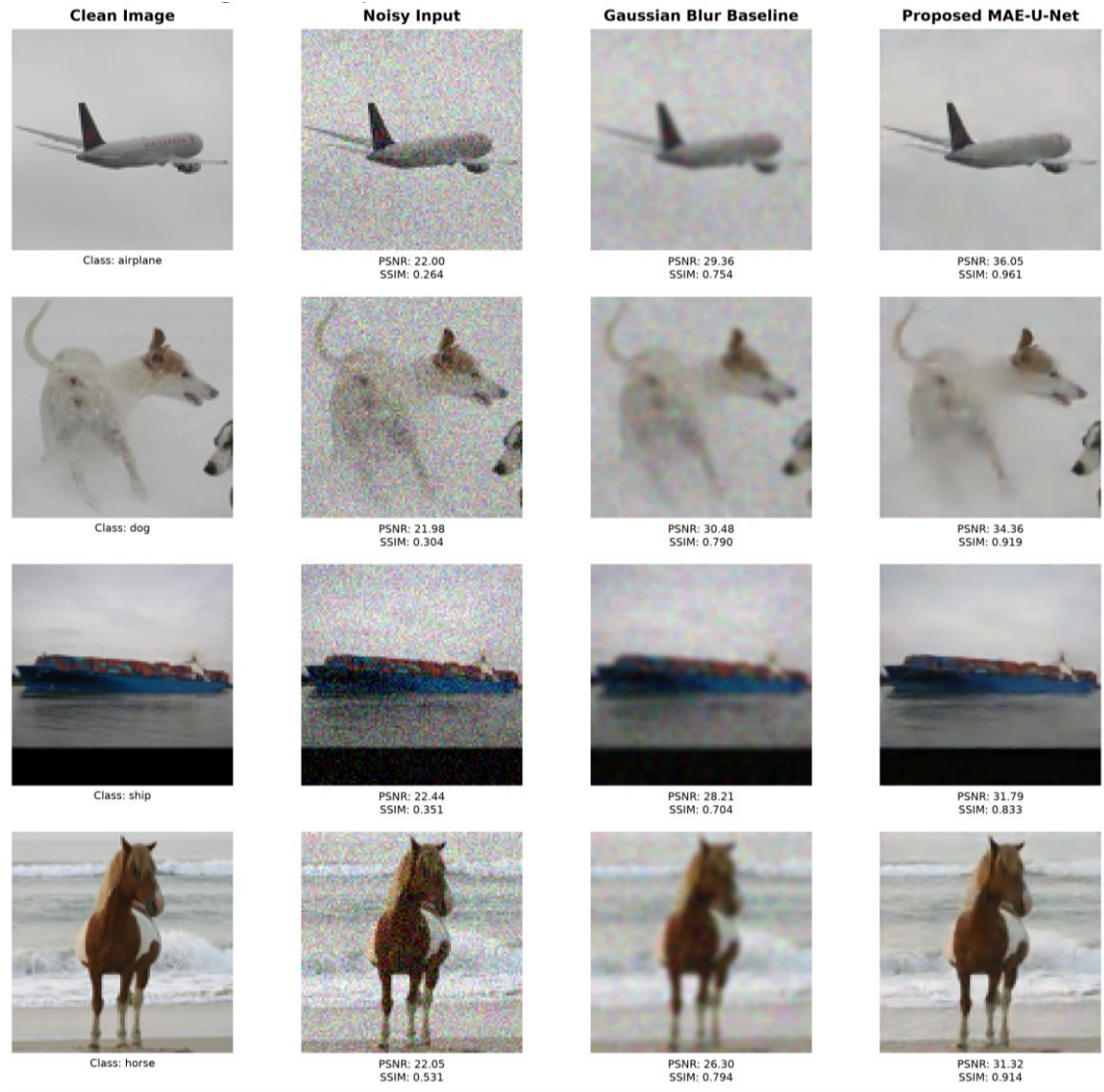


Figure 5 Comparison with traditional Gaussian blur baseline

Figure 5 shows clean images, noisy inputs, outputs from the Gaussian blur baseline and outputs from the proposed MAE-U-Net. While the Gaussian blur baseline reduces visible noise, the resulting images are often smoother and less detailed. In contrast, the proposed model produces cleaner outputs while preserving more structural information.

The difference between Gaussian blur and the proposed model is significant. Gaussian blur performs the same mathematical operation on all image regions. It does not understand image content and cannot distinguish between noise and important texture. The proposed model, however, learns from image data. Through self-supervised reconstruction, the encoder learns visual representations of natural image structures. The decoder then uses these representations to reconstruct clearer images.

This comparison demonstrates that the proposed model is capable of more than just simple smoothing. It learns a data-driven enhancement function that is more effective than a fixed

classical filter. This supports the report's main argument that artificial intelligence and self-supervised representation learning can address image enhancement more effectively than traditional non-learning methods.

4.6 Quantitative Results

The quantitative evaluation confirms the visual improvements observed in the qualitative results. The proposed MAE-U-Net model was compared with noisy input images and Gaussian blur outputs using PSNR and SSIM.

Method	Average PSNR (dB)	Average SSIM	Interpretation
Noisy input	22.27	0.6031	Degraded image with visible Gaussian noise
Gaussian blur baseline	24.22	0.7590	Noise is reduced, but structural details may be smoothed
Proposed MAE-U-Net	28.89	0.8875	Best reconstruction quality and structural preservation

Table 10 Quantitative Comparison of Enhancement Results

The proposed method achieved an average PSNR of 28.89 dB, compared to 22.27 dB for the noisy input. This represents a 6.62 dB improvement over the noisy input. The proposed model also outperformed the Gaussian blur baseline, achieving 24.22 dB. The PSNR improvement over Gaussian blur was 4.67 dB. These results demonstrate that the proposed model can reconstruct images more accurately at the pixel level.

The SSIM results also demonstrate significant improvement. While the noisy input achieved an average SSIM of 0.6031, the Gaussian blur baseline improved this to 0.7590. The proposed model achieved the highest SSIM value of 0.8875. This suggests that the model preserves structural information more effectively than the noisy input and the Gaussian blur baseline.

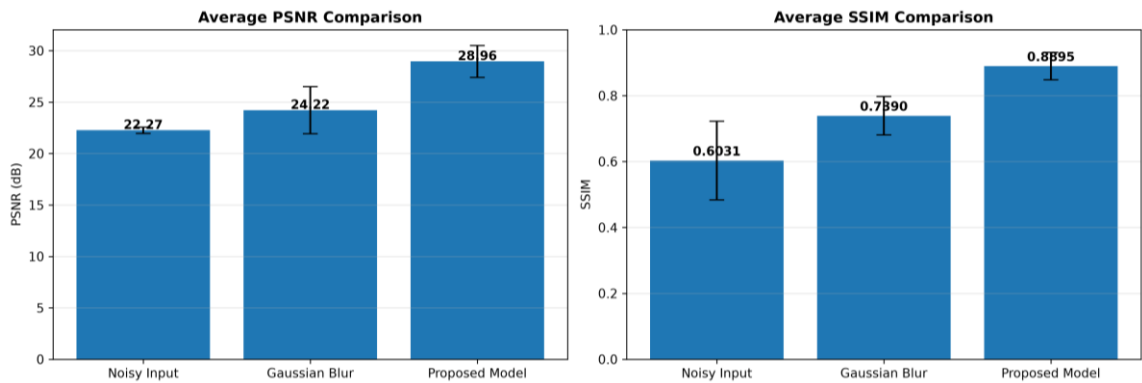


Figure 6 Quantitative Evaluation of the Proposed Enhancement Model

Figure 6 visually presents the PSNR and SSIM comparison. The proposed model achieves the highest values in both metrics. This confirms that the improvement is not only visual, but also supported by objective image quality measurements.

The quantitative results demonstrate three important points. Firstly, the model successfully learnt to reduce Gaussian noise. Secondly, the model preserves image structure more effectively than Gaussian blur. Thirdly, the self-supervised reconstruction strategy was effective despite class labels not being used during training. These findings support the report's main goal.

4.7 Noise-level generalisation

To further evaluate the robustness of the proposed model, experiments were conducted at different levels of Gaussian noise. This analysis examines whether the learned representations can generalise beyond the main training noise level. Although the model was primarily trained using $\sigma = 0.08$, it was evaluated under several noise levels.

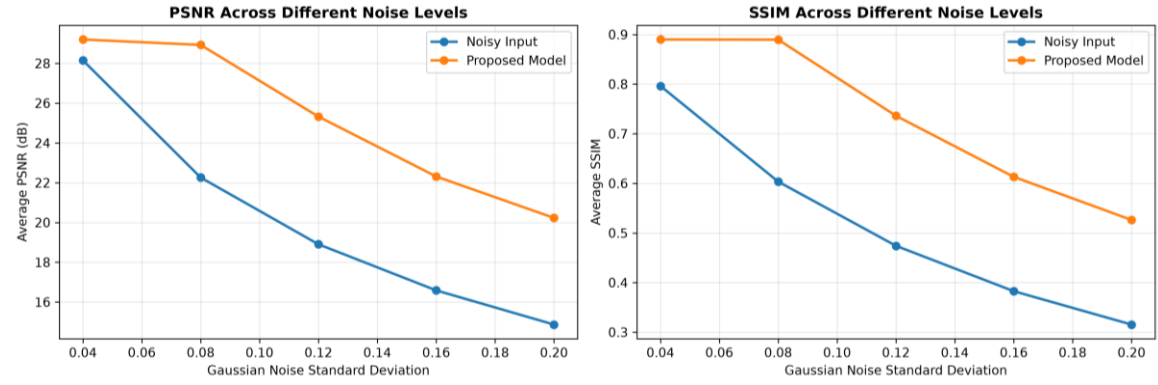


Figure 7 Generalisation of Learned Representations Across Noise Levels

Figure 7 shows the PSNR and SSIM performance of the proposed model across different standard deviations of Gaussian noise. The results demonstrate that the proposed model consistently outperforms noisy inputs across the tested noise levels. As the noise level increases, the quality of the noisy input decreases, as expected. However, the proposed model still produces improved outputs because it has learned useful denoising and reconstruction representations.

This is an important result because it shows that the model is not merely memorising one fixed noise pattern. Instead, it learns general image restoration behaviour through the self-supervised reconstruction task. The encoder learns features related to image structure, and

the decoder uses these features to reconstruct enhanced images. This enables the model to handle different levels of degradation.

At very low noise levels, the difference between the noisy input and the enhanced output may be smaller because the input image is already similar to the clean image. At higher noise levels, however, the improvement becomes more noticeable because the model has more degradation to remove. This trend is reasonable and supports the stability of the proposed method.

4.8 Visualisation of Learned Representations

As this report focuses on self-supervised representation learning, it is important to analyse both the enhanced outputs and the learned encoder representations. To visualise the learned feature space, encoder features were extracted from test images and projected into two dimensions using principal component analysis.

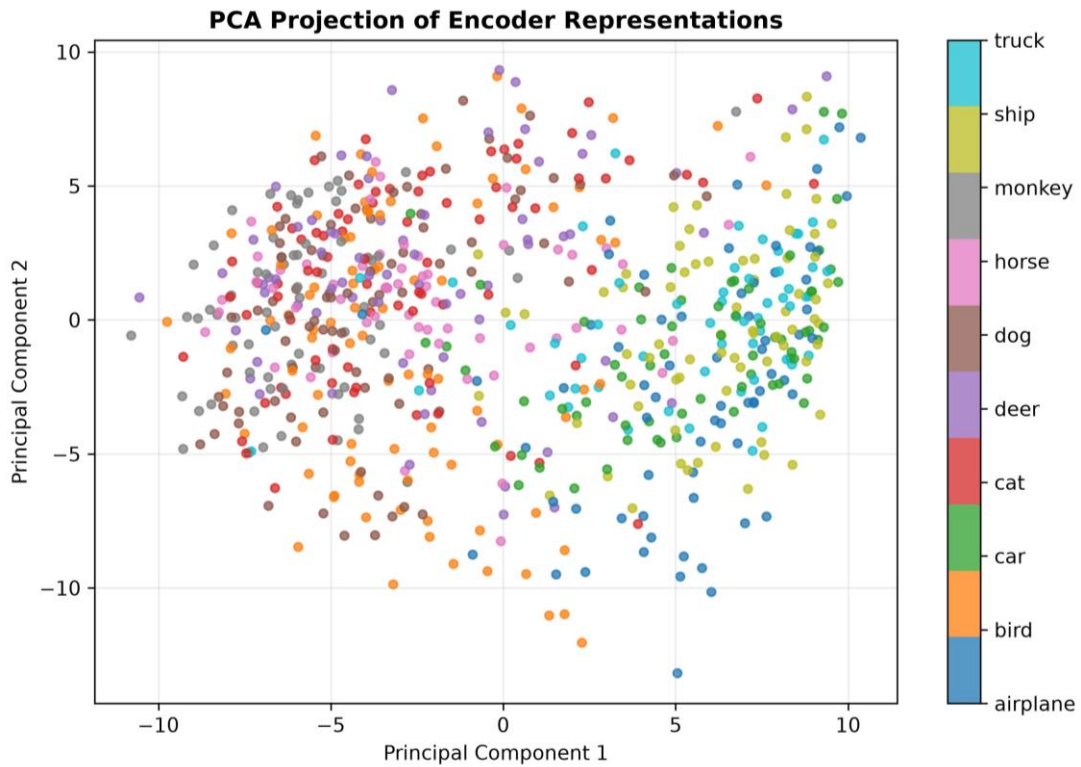


Figure 8 Visualisation of Self-Supervised Learned Representations

Figure 8 shows the PCA visualisation of the learned encoder representations. Each point represents an image feature extracted by the encoder and different colours correspond to different STL-10 image categories. Although class labels were not used during training, the

feature distribution demonstrates that the encoder has learned to recognise structured visual representations in the image data.

However, the PCA visualisation should not be interpreted as a classification result. This figure does not aim to demonstrate perfect class separation. Instead, it demonstrates that the encoder organises image features in a meaningful representation space. This supports the self-supervised learning aspect of the report. Although the model learns from reconstruction rather than labels, the learned features still reflect visual differences among image samples.

This representation analysis strengthens the report by connecting the experimental results to the assigned topic. Not only is the method a denoising model, it is also a self-supervised representation learning model applied to image enhancement.

4.9 Discussion of main findings

The experimental results demonstrate the effectiveness of the proposed self-supervised MAE-U-Net model for image enhancement. The model improves both visual quality and quantitative metrics. The enhanced images are cleaner than the noisy inputs and preserve more structure than the Gaussian blur baseline.

The strongest evidence comes from the PSNR and SSIM results. The proposed model increases PSNR by 6.62 dB, improving it from 22.27 to 28.89, and increases SSIM by 0.2844, improving it from 0.6031 to 0.8875. These improvements suggest that the model successfully reduces noise while preserving structural similarity. The improvement over Gaussian blur also confirms that the model is not merely smoothing the image. Instead, it learns a more adaptive enhancement function from the data.

The training loss curve also supports the effectiveness of the method. The consistent decrease in loss over 20 epochs shows that the model successfully learned the reconstruction task. As the model was trained without class labels, this result confirms that the self-supervised corruption strategy provided a meaningful learning signal.

Qualitative results demonstrate the benefits of using a U-Net architecture. Compared with simple autoencoder-based reconstruction, U-Net's skip connections help to preserve spatial details and reduce excessive blurring. This is important for image enhancement, as the aim is to reduce noise while retaining important visual information.

The noise-level generalisation experiment shows that the learned model can handle different levels of Gaussian noise. This suggests that the encoder has learned useful image representations rather than simply memorising a single corrupted pattern. PCA

visualisation also supports this idea, showing that the encoder features form a structured representation space.

Experimental component	Main observation	Meaning
Training loss curve	Loss decreases steadily over 20 epochs	Model learns the self-supervised reconstruction task
Qualitative results	Enhanced outputs are cleaner than noisy inputs	Model improves visual quality
Baseline comparison	Proposed model is clearer than Gaussian blur	Deep learning model preserves structure better
PSNR evaluation	PSNR improves from 22.27 dB to 28.89 dB	Pixel-level reconstruction accuracy improves
SSIM evaluation	SSIM improves from 0.6031 to 0.8875	Structural similarity improves
Noise generalization	Model remains effective across noise levels	Learned representations are reasonably robust
PCA visualization	Encoder features show structured distribution	Model learns useful visual representations

Table 11 Summary of Experimental Findings

4.10 Limitations

Although the proposed method produced strong results, there are still some limitations. Firstly, the experiment used synthetic Gaussian noise. Real-world image degradation may include more complex patterns of noise, such as low-light noise, motion blur, compression artefacts or mixed degradation. Therefore, further testing is required to assess performance on real-world degraded images.

Secondly, the model was trained and evaluated using STL-10 images that had been resized to 96×96 pixels. While this resolution is suitable for a course project and provides clear visual results, higher-resolution datasets may be required for practical image enhancement applications. Training on larger images could improve detail preservation, but would require more computational resources.

Thirdly, the proposed method uses a U-Net-based architecture rather than a full transformer-based masked autoencoder. While the U-Net model provides good results and preserves spatial details, transformer-based models may learn stronger global representations. Future work could involve comparing the current MAE-U-Net with transformer-based masked autoencoders.

Fourthly, the evaluation primarily relied on PSNR and SSIM. While these metrics are standard and useful, they may not fully represent human visual perception. Future studies

could therefore include perceptual metrics or human visual assessment to provide a more comprehensive evaluation.

5. Conclusion and future work

5.1 Conclusion

This report presented a self-supervised representation learning approach for image enhancement, based on a Masked Autoencoder-inspired U-Net model. The study's primary objective was to demonstrate how a deep learning model can learn useful visual representations from corrupted image data and apply them to enhance image quality. The proposed method was designed in line with the topic 'Self-supervised representation learning for image enhancement', whereby the model learns without using class labels and performs image enhancement through reconstruction.

Gaussian noise and random patch masking were used to create self-supervised corrupted image views for the study. These corruptions enabled the model to learn from the image's underlying structure. The encoder learnt visual representations from noisy and partially masked inputs, while the U-Net decoder reconstructed enhanced images using high- and low-level features, as well as spatial information. Using U-Net skip connections was important because image enhancement requires noise reduction and preservation of edges, textures and object boundaries.

Experiments were conducted on the STL-10 dataset using 96×96 RGB images. The model was trained for 20 epochs with Gaussian noise at $\sigma = 0.08$, a patch size of 8×8 pixels and a random masking ratio of 30%. The proposed method was evaluated using PSNR and SSIM, which are standard metrics for image restoration and enhancement. A Gaussian blur method was also employed as a traditional baseline for comparison.

The experimental results showed that the proposed model achieved significant performance enhancement. The average PSNR increased from 22.27 dB for the noisy inputs to 28.89 dB for the enhanced outputs, representing a gain of 6.62 dB. The average SSIM increased from 0.6031 to 0.8875, indicating that the enhanced outputs preserved more structural information than the original noisy images. Compared with the Gaussian blur baseline, the proposed MAE-U-Net model achieved higher PSNR and SSIM values, confirming that it performed better than simple traditional smoothing.

The qualitative results supported the quantitative findings. The enhanced images appeared cleaner and more visually stable than the noisy inputs. Unlike Gaussian blur, which reduces noise but also smooths details, the proposed model preserves more object structure and image clarity. The training loss curve showed stable convergence over 20 epochs, indicating

that the model had successfully learned the reconstruction task. Furthermore, the noise-level generalisation experiment showed that the model could handle different levels of Gaussian noise. Additionally, PCA visualisation of encoder features revealed that the model had learned structured visual representations from the self-supervised task.

Overall, these results demonstrate the effectiveness of applying self-supervised representation learning to image enhancement. Despite being trained without class labels, the model learned useful visual features for reconstructing cleaner images. This confirms that self-supervised learning is a valuable approach when labelled data or paired clean-degraded images are limited. The proposed MAE-U-Net framework therefore provides a practical, deep-learning-based solution for image enhancement in digital image processing.

5.2 Limitations

Although the proposed method produced strong results, several limitations should be considered. Firstly, the experiment primarily employed synthetic Gaussian noise. Real-world image degradation is often more complex, potentially including sensor noise, low-light noise, motion blur, compression artefacts and mixed distortions. Therefore, further investigation is needed into the performance of the model on real degraded images.

Secondly, the study used the STL-10 dataset, in which the images were resized to 96×96 pixels. While this resolution was suitable for the current experiment and produced clear visual results, real-world image enhancement often requires higher-resolution images. Training and testing on larger images would improve practical applicability but would also require more computational resources.

Thirdly, the proposed model used a U-Net-based architecture rather than a full transformer-based masked autoencoder. While the U-Net design was effective for preserving spatial details, transformer-based models may capture stronger global image relationships. Therefore, further improvement of the current work could be achieved by comparing CNN- and transformer-based self-supervised enhancement models.

Fourthly, the evaluation primarily employed PSNR and SSIM. While these metrics are standard and useful, they may not fully reflect human visual perception. Some images may have good PSNR and SSIM values yet appear slightly over-smoothed. Therefore, future work should also consider perceptual metrics or human visual evaluation to more completely assess visual quality.

5.3 Future work

This study can be extended in several ways in future. Firstly, the model could be tested using real-world noisy image datasets rather than synthetic Gaussian noise only. This would provide a more realistic evaluation and help determine whether the model can handle complex degradation patterns.

Secondly, the method could be improved by using larger, higher-resolution datasets. Training on high-resolution images may enable the model to learn finer textures and produce sharper enhanced outputs. This would be particularly beneficial for applications such as medical imaging, remote sensing, surveillance and natural image restoration.

Thirdly, transformer-based masked autoencoders could be explored as an extension of the current MAE-U-Net framework. Vision transformers can capture long-range dependencies and global image context, which could enhance representation learning. Comparing CNN-based U-Net models with transformer-based masked autoencoders in future would provide deeper insight into the role of architecture in self-supervised image enhancement.

Fourthly, adding additional loss functions could improve perceptual quality. Perceptual loss, edge-preserving loss, and adversarial loss, for example, may help to reduce over-smoothing and produce sharper outputs. These losses could be combined with PSNR- and SSIM-based objectives to strike a balance between numerical accuracy and visual quality.

Finally, this method can be adapted for use in other image restoration tasks, such as low-light enhancement, deblurring, super-resolution and image inpainting. As the proposed model learns visual representations from corrupted images, the same self-supervised learning concept can be applied to various digital image processing issues.

In conclusion, this report demonstrates that a self-supervised MAE-U-Net model can effectively enhance noisy images without the need for class labels. Improvements in PSNR, SSIM, qualitative results, baseline comparison and representation visualisation confirm the effectiveness and relevance of the proposed method to AI-based digital image processing. Further improvement and testing on more complex datasets would further demonstrate the strong potential of self-supervised representation learning for practical image enhancement applications.

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